

Data Dictionary

Student Grade Management System

Table 1: students

Database: student_db

Description: Stores one record per student, including identifying details.

Column	Data Type	Field Size	Constraint	Nullable	Java Mapping	Description
student_id	SERIAL (int)	4	PRIMARY KEY, Auto-increment	NO	int	Unique identifier for each student. Auto-generated by PostgreSQL. Never set manually on insert.
registration_number	VARCHAR(20)	20	NOT NULL, UNIQUE	NO	String	School-issued registration number used to look up and identify a student. Maximum 20 characters. Must be unique across all students.
name	VARCHAR(100)	100	NOT NULL	NO	String	Full name of the student. Maximum 100 characters.
created_at	TIMESTAMP	8	NOT NULL, DEFAULT NOW()	NO	Timestamp / String	Date and time the student record was created. Automatically set by PostgreSQL when the row is inserted.

Table 2: subjects

Database: student_db

Description: Stores the list of subjects taught. Allows new subjects to be added without changing the table structure.

Column	Data Type	Field Size	Constraint	Nullable	Java Mapping	Description
subject_id	SERIAL (int)	4	PRIMARY KEY, Auto-increment	NO	int	Unique identifier for each subject. Auto-generated by PostgreSQL.
subject_name	VARCHAR(50)	50	NOT NULL, UNIQUE	NO	String	Name of the subject, e.g. Math, English, Science, Social Studies. UNIQUE ensures no duplicate subject names.

Table 3: scores

Database: student_db

Description: Links students to subjects and stores their score for each. This is the junction table that connects students and subjects.

Column	Data Type	Field Size	Constraint	Nullable	Java Mapping	Description
score_id	SERIAL (int)	4	PRIMARY KEY, Auto-increment	NO	int	Unique identifier for each score record. Auto-generated by PostgreSQL.
student_id	INTEGER	4	NOT NULL, FOREIGN KEY -> students(student_id), ON DELETE CASCADE	NO	int	References the student the score belongs to. If the student is deleted, their scores are deleted too.
subject_id	INTEGER	4	NOT NULL, FOREIGN KEY -> subjects(subject_id), ON DELETE CASCADE	NO	int	References the subject the score is for. If the subject is deleted, related scores are deleted too.
score	NUMERIC(5,2)	5,2	NOT NULL, CHECK (score >= 0 AND score <= 100)	NO	double	The student's score for the subject, stored as an exact decimal with up to 2 decimal places. Must be between 0 and 100.

Relationships Between Tables

students — scores — subjects

- One student can have many scores (one per subject).
- One subject can have many scores (one per student).
- scores is the junction table that sits between them.
- This is a many-to-many relationship between students and subjects, resolved through scores.